

AI-Enhanced Geospatial Tourism Recommendation System

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Abstract—Tourism planning has become increasingly complex due to the vast amount of travel-related information available across digital platforms. Travelers often rely on multiple online sources to identify suitable destinations, explore nearby attractions, and determine the best time to visit a location. However, many existing travel recommendation systems primarily provide generic suggestions based on popularity or ratings, without considering important factors such as seasonal suitability, user preferences, and geographic proximity between tourist destinations. As a result, travelers frequently experience difficulty in identifying meaningful and personalized travel options, which can lead to inefficient trip planning and unsatisfactory travel experiences.

To address these challenges, this research proposes an AI-Enhanced Geospatial Tourism Recommendation System that integrates user preference analysis, seasonal filtering, geospatial distance computation, and AI-generated travel insights to provide personalized tourism recommendations. The proposed system allows users to specify their preferred travel month and destination category, such as beaches, heritage sites, nature locations, adventure destinations, or wildlife attractions. Based on these inputs, the system filters suitable destinations from a structured tourism dataset containing information such as location coordinates, descriptive details, and recommended travel seasons.

The system applies geospatial analysis using the Haversine formula to calculate distances between tourist locations based on their latitude and longitude coordinates. This enables the system to identify nearby attractions that travelers can explore within a defined geographic radius. Additionally, the system integrates a generative artificial intelligence module that produces travel insights, including recommended travel duration, suggested activities, and practical travel tips for selected destinations. These insights enhance the decision-making process by providing contextual information that supports effective travel planning.

The proposed system is implemented as a full-stack web application using modern web technologies. The frontend interface is developed using React and TypeScript, while the backend is implemented using Node.js and Express.js with a lightweight SQLite database. An interactive map visualization using the Leaflet library enables users to explore tourist destinations spatially and view nearby attractions on a geographic map.

Experimental usage demonstrates that integrating user preference filtering, seasonal analysis, geospatial proximity calculations, and AI-generated insights significantly improves the relevance and usability of tourism recommendations compared to traditional search-based travel platforms. The

system therefore provides a practical and user-friendly solution for intelligent tourism planning.

Keywords —*Tourism Recommendation System, Geospatial Analysis, Artificial Intelligence, Travel Planning, Nearby Attraction Recommendation, Location-Based Services, Map Visualization.*

I. INTRODUCTION

Tourism is one of the fastest growing global industries and plays a significant role in economic development, cultural exchange, and international connectivity. With the rapid growth of digital technologies and internet accessibility, travelers increasingly depend on online platforms to explore destinations, gather travel information, and plan their journeys. Modern travel websites and mobile applications provide a wide range of tourism-related information such as destination descriptions, user reviews, accommodation options, and travel guides. Although these platforms offer large volumes of travel information, they often overwhelm users with excessive choices and lack personalized guidance tailored to individual traveler needs.

One of the major limitations of existing tourism platforms is the lack of intelligent recommendation mechanisms that consider important contextual factors such as **user interests, seasonal suitability, and geographic proximity between tourist locations**. Many traditional travel platforms primarily rely on popularity rankings, user ratings, or general search queries to recommend destinations. While these approaches may help users discover well-known tourist attractions, they frequently fail to provide recommendations that align with the traveler's personal preferences or travel timing. For example, a traveler searching for destinations during a particular month may receive recommendations that are not suitable due to unfavorable weather conditions or seasonal limitations.

Another important challenge in travel planning is identifying **nearby attractions and geographically related locations**. Travelers often wish to explore multiple places during a single trip, especially when destinations are located within a short geographic distance from each other. However, many travel platforms require users to manually search for nearby attractions, which can make travel planning time-consuming and inefficient. Without proper geospatial analysis, users may miss opportunities to explore nearby tourist destinations that could significantly enhance their travel experience.

Recent advancements in **artificial intelligence (AI) and geospatial technologies** have opened new possibilities for improving tourism recommendation systems. Artificial intelligence techniques can analyze user preferences and generate intelligent suggestions, while geographic information systems enable the analysis of spatial relationships between locations. Integrating these technologies allows travel platforms to provide more personalized and context-aware recommendations that improve the overall travel planning experience.

Another key factor influencing travel decisions is the **best time to visit a destination**. Many tourist locations offer optimal experiences only during certain seasons due to climate conditions, festivals, or environmental factors. However, many existing tourism recommendation systems do not effectively incorporate seasonal information into their recommendation processes. As a result, travelers may receive suggestions that are not appropriate for their selected travel period.

To address these challenges, this research proposes an **AI-Enhanced Geospatial Tourism Recommendation System** that integrates user preference filtering, seasonal suitability analysis, geospatial distance calculation, and AI-generated travel insights to provide meaningful and personalized tourism recommendations. The system allows users to select their preferred travel month and category of interest, such as beaches, heritage sites, nature destinations, adventure locations, or wildlife attractions. Based on these inputs, the system filters suitable destinations from a tourism dataset and provides detailed information about each location.

In addition to preference-based filtering, the system performs **geospatial distance analysis using the Haversine formula** to calculate the geographic distance between tourist destinations based on latitude and longitude coordinates. This enables the system to recommend nearby attractions that travelers can explore within a defined geographic range. Furthermore, the system integrates a generative artificial intelligence module that produces travel insights such as recommended travel duration, suggested activities, and practical travel tips for selected destinations.

The proposed system is implemented as a **full-stack web application** that combines modern web technologies with geospatial analysis. The frontend interface provides an interactive and user-friendly experience that allows travelers to explore destinations visually through map-based visualization. The system also includes features such as travel history tracking, nearby attraction discovery, and accommodation recommendations.

The primary objective of this research is to develop an intelligent tourism recommendation platform that simplifies travel planning and provides context-aware destination suggestions. By integrating user preferences, seasonal filtering, geospatial proximity analysis, and AI-generated travel insights, the proposed system aims to enhance the accuracy and usability of tourism recommendation services while improving the overall travel experience for users.

II. RELATED WORK

Tourism recommendation systems have gained significant attention in recent years due to the rapid growth of digital travel platforms and the increasing demand for personalized travel experiences. Researchers have explored various techniques to improve tourism recommendation systems, including collaborative filtering, content-based filtering, geographic information systems, and artificial intelligence-based recommendation models.

One of the earliest approaches used in tourism recommendation systems is **collaborative filtering**, which recommends tourist destinations based on the preferences and behavior of similar users. Wang [2] proposed a tourism recommendation model based on collaborative filtering that analyzes user ratings and historical travel behavior to identify patterns and generate destination suggestions. Collaborative filtering techniques are widely used in many recommendation platforms; however, they often suffer from the **cold-start problem**, where the system cannot provide accurate recommendations for new users or newly added destinations due to the absence of sufficient historical data.

Content-based recommendation systems represent another widely studied approach in tourism applications. These systems analyze attributes associated with tourist destinations, such as textual descriptions, categories, and keywords, and match them with user preferences. Băcilă et al. [10] explored the use of content analysis techniques to understand tourism promotion through multimedia content and social media platforms. While content-based systems can improve recommendation relevance, many of them do not consider geographic relationships between tourist destinations. As a result, recommendations may include destinations that are geographically distant from each other, making travel planning inefficient.

Geographic Information Systems (GIS) have also been applied to tourism recommendation platforms to improve spatial analysis of tourist locations. Tang and Painho [11] investigated the relationship between spatial data and user-generated content in tourism applications. GIS-based approaches use geographic coordinates such as latitude and longitude to determine distances between destinations and identify nearby attractions. These systems can support location-based recommendations and route planning. However, many GIS-based systems focus primarily on spatial analysis without integrating user preferences or seasonal suitability into the recommendation process.

Recent research has also explored the use of **artificial intelligence and machine learning techniques** to enhance tourism recommendation systems. Ruan et al. [8] proposed a recommendation framework that analyzes user behavior and emotional features to generate personalized suggestions. Artificial intelligence methods enable tourism platforms to process large amounts of travel-related information and extract meaningful insights from user interactions. Nevertheless, many AI-based tourism recommendation systems require complex computational models and extensive datasets, which may increase implementation complexity and computational cost.

Other studies have investigated the impact of online travel

platforms on traveler decision-making behavior. Abashidze [1] examined how online platforms influence travel decisions and user behavior patterns, highlighting the importance of intelligent recommendation systems that provide relevant travel information. Similarly, Surbakti et al. [3] discussed the role of artificial intelligence in improving operational efficiency in online travel platforms by transforming raw travel data into actionable recommendations.

In addition to recommendation techniques, some research has focused on improving tourism experiences through the integration of emerging technologies. Farid et al. [5] explored the use of Internet of Things (IoT) technologies to create smart tourism environments where tourists can access real-time information about destinations. These technologies can enhance travel experiences by providing dynamic and location-aware information.

Despite these advancements, many existing tourism recommendation systems still lack an integrated approach that simultaneously considers **user preferences, seasonal travel suitability, and geographic proximity between destinations**. Some systems emphasize popularity-based recommendations, while others focus solely on spatial analysis or textual content matching. As a result, travelers may receive recommendations that are either irrelevant to their interests or impractical for their travel schedules.

The tourism recommendation system proposed in this research addresses these limitations by integrating **preference-based filtering, seasonal analysis, geospatial distance calculation, and AI-generated travel insights** within a single platform. By combining these techniques, the system provides personalized and context-aware tourism recommendations while maintaining a lightweight web-based architecture suitable for real-world travel applications.

III. DATA COLLECTION & PREPROCESSING

The effectiveness of any tourism recommendation system largely depends on the quality and structure of the dataset used for generating recommendations. In this research, a structured tourism dataset was developed to support intelligent travel recommendations across multiple regions in India. The dataset contains detailed information about tourist destinations, including geographic coordinates, destination categories, seasonal suitability, descriptive details, and accommodation information.

The primary goal of the dataset is to enable the system to perform **preference-based filtering and geospatial analysis** for tourism recommendations. Each destination entry in the dataset contains several attributes that allow the system to identify suitable locations based on user inputs. These attributes include the destination name, city, state, country, latitude and longitude coordinates, category of attraction, descriptive information, and recommended travel months.

The dataset includes a diverse collection of tourist destinations distributed across different states of India. These destinations are categorized into several tourism categories such as **beaches, heritage sites, nature locations, wildlife attractions, and adventure destinations**. Categorization plays an important role in the recommendation process

because it allows the system to match user interests with relevant destinations. For example, a user interested in nature tourism can receive recommendations for national parks, hill stations, and wildlife sanctuaries.

In addition to destination information, the dataset also contains **accommodation details** associated with tourist locations. These records include information about hotels, resorts, hostels, and guest houses available near each destination. Accommodation data includes attributes such as accommodation name, accommodation type, price range, distance from the destination center, image links, and external booking links. This information allows the system to provide users with accommodation options that complement their travel plans.

Geographic information plays a crucial role in the dataset because it enables the system to perform **location-based analysis**. Each tourist destination contains latitude and longitude coordinates that allow the system to calculate distances between locations using geospatial algorithms. These geographic coordinates are used by the system to identify nearby tourist attractions that can be visited within a specific travel radius.

Before integrating the dataset into the tourism recommendation system, several preprocessing steps were performed to ensure data accuracy and consistency. Data preprocessing is essential for improving the reliability of the recommendation process and reducing potential errors during data retrieval.

The first preprocessing step involved **removing duplicate destination entries** that could appear due to repeated data collection from multiple tourism sources. Duplicate removal ensures that the recommendation system does not generate redundant suggestions for the same location.

The second preprocessing step involved **verifying geographic coordinates** for each destination. Incorrect or inconsistent latitude and longitude values can affect distance calculations and lead to inaccurate nearby attraction recommendations. Therefore, geographic coordinates were validated to ensure accurate geospatial analysis.

The third preprocessing step focused on **standardizing categorical attributes**, including destination categories and travel months. Standardization ensures that category labels such as "Nature," "Wildlife," or "Adventure" are consistently represented across the dataset. Similarly, travel months were formatted in a consistent structure to support efficient seasonal filtering.

Another preprocessing task involved ensuring that **textual descriptions of destinations** were complete and properly formatted. These descriptions provide contextual information about each tourist location and improve the overall user experience by offering meaningful insights about destinations.

Finally, the processed dataset was stored in a **SQLite relational database**, which provides a lightweight and efficient storage solution for the tourism recommendation system. SQLite allows the application to quickly retrieve destination information, filter locations based on user

preferences, and access geographic coordinates required for distance calculations.

By organizing tourism data into a structured database with verified geographic coordinates and standardized attributes, the system ensures reliable recommendation results. This dataset forms the foundation of the proposed tourism recommendation platform and supports the integration of preference filtering, seasonal analysis, geospatial proximity detection, and AI-generated travel insights.

IV. PROPOSED METHODOLOGY.

The proposed **AI-Enhanced Geospatial Tourism Recommendation System** is designed to provide intelligent travel recommendations by combining user preference analysis, seasonal suitability filtering, geospatial distance calculation, and AI-generated travel insights. The system follows a multi-stage workflow that processes user inputs and tourism data to generate meaningful travel suggestions.

The overall objective of the proposed methodology is to simplify travel planning by recommending tourist destinations that match the user's interests while also considering travel timing and geographic proximity between locations. The architecture of the system consists of several interconnected components including the user interface, recommendation engine, geospatial analysis module, database management system, and AI insight generator.

A. User Input and Preference Selection

The recommendation process begins when the user interacts with the web application interface. The system allows users to select their **preferred travel month** and **destination category** based on their interests. The available destination categories include beaches, heritage locations, nature destinations, adventure tourism, and wildlife attractions.

The selected inputs serve as the primary parameters for the recommendation engine. These parameters help the system filter relevant destinations from the tourism dataset stored in the database. By allowing users to specify their travel preferences, the system ensures that the generated recommendations align with the user's interests and travel schedule.

B. Seasonal Filtering of Destinations

Seasonal suitability is an important factor in tourism planning because many destinations offer the best travel experiences only during specific months of the year. For example, hill stations may be most attractive during summer months, while beach destinations may be more suitable during winter seasons.

To address this requirement, the system performs **seasonal filtering** by analyzing the travel month selected by the user. Each destination in the dataset contains information about the recommended months for visiting the location. During the filtering process, the system compares the user-selected month with the recommended travel months stored in the dataset. Only destinations that match the selected travel period are considered during the recommendation process.

This filtering step helps eliminate destinations that may not provide optimal travel experiences due to seasonal conditions.

C. Category-Based Destination Filtering

After seasonal filtering, the system applies **category-based filtering** to identify destinations that match the user's travel interests. Each destination in the dataset is associated with a specific category such as nature tourism, heritage tourism, beach tourism, adventure tourism, or wildlife tourism.

The system compares the selected category with the category attribute stored in the destination records. Destinations that match the selected category are retained for further processing. This filtering mechanism ensures that users receive recommendations that align with their travel preferences.

By combining seasonal filtering with category filtering, the system significantly reduces the number of irrelevant destinations and improves recommendation accuracy.

D. Geospatial Distance Analysis

Once suitable destinations are identified through filtering, the system performs **geospatial distance analysis** to determine nearby tourist attractions. Geographic proximity plays a critical role in travel planning because travelers often prefer to visit multiple nearby destinations during a single trip.

The system uses the **Haversine formula** to calculate the distance between two geographic coordinates based on latitude and longitude values. The formula determines the shortest distance between two points on the Earth's surface.

The Haversine distance formula is expressed as:

$$d = 2R * \arcsin(\sqrt{(\sin^2((\phi_2 - \phi_1)/2) + \cos(\phi_1)\cos(\phi_2)\sin^2((\lambda_2 - \lambda_1)/2))})$$

Where:

- d represents the distance between two locations
- R represents the radius of the Earth (approximately 6371 km)
- ϕ_1 and ϕ_2 represent the latitude values of two locations
- λ_1 and λ_2 represent the longitude values of two locations

Using this formula, the system calculates the distance between the selected destination and other tourist locations stored in the dataset. Destinations that fall within a predefined radius are considered **nearby attractions** and are recommended to the user.

This geospatial analysis helps travelers discover additional places of interest that are geographically close to their primary destination.

E. AI-Based Travel Insights Generation

To further enhance the travel planning experience, the system integrates an **AI-powered travel insight module**. This module uses a generative artificial intelligence service to

produce useful travel information related to selected destinations.

The AI module generates insights such as:

- Recommended travel duration for the destination
- Suggested tourist activities
- Practical travel tips and advice

These AI-generated insights provide contextual information that helps travelers better understand the destination and plan their trips more effectively.

F. Interactive Map Visualization

The final stage of the recommendation process involves presenting the results through an interactive user interface. The system displays recommended destinations along with nearby attractions and accommodation options.

An **interactive map visualization** is implemented using the Leaflet mapping library. The map displays the geographic location of the selected destination using markers based on latitude and longitude coordinates. Users can interact with the map to explore nearby tourist locations visually.

Map-based visualization improves user engagement and helps travelers understand the spatial relationships between destinations and nearby attractions.

G. Overall Recommendation Workflow

The overall workflow of the proposed system can be summarized in the following steps:

1. User selects preferred travel month and destination category.
2. System performs seasonal filtering of tourist destinations.
3. Category-based filtering identifies destinations matching user interests.
4. Geospatial analysis calculates distances between destinations.
5. Nearby attractions within a defined radius are identified.
6. AI module generates travel insights for selected destinations.
7. Results are displayed through an interactive web interface with map visualization.

By integrating these components, the proposed system provides a comprehensive and intelligent tourism recommendation platform that improves travel planning efficiency and enhances the overall user experience.

V. IMPLEMENTATION

The proposed **AI-Enhanced Geospatial Tourism Recommendation System** was implemented as a full-stack web application designed to provide an interactive and user-friendly tourism planning platform. The implementation integrates modern web development technologies with geospatial analysis techniques and artificial intelligence services to generate intelligent tourism recommendations.

The system architecture consists of three primary components: the **frontend user interface**, the **backend server and API services**, and the **database management system**. These components work together to process user inputs, retrieve tourism data, perform recommendation calculations, and present results through an interactive interface.

A. Frontend Implementation

The frontend of the application was developed using **React with TypeScript**, which enables the creation of modular, reusable, and scalable user interface components. React provides a component-based architecture that allows different parts of the interface such as navigation bars, destination cards, maps, and recommendation panels to be developed independently and integrated into a cohesive application.

To enhance the visual design and responsiveness of the application, **Tailwind CSS** was used as the primary styling framework. Tailwind provides utility-based CSS classes that simplify the development of modern and responsive layouts. The use of Tailwind CSS allowed the interface to maintain a consistent visual appearance across different screen sizes and devices.

Navigation between different sections of the application is handled using **React Router**, which enables seamless page transitions without requiring full page reloads. Key pages in the system include:

- Welcome and authentication pages
- User dashboard
- Travel planner interface
- Destination details page
- Nearby attractions page
- Accommodation recommendations page

The frontend communicates with the backend server using **HTTP API requests** to retrieve recommendation data, user history, and destination details.

B. Backend Implementation

The backend server was implemented using **Node.js with Express.js**, which provides a lightweight and efficient environment for building web application APIs. The Express framework handles routing, request processing, and communication between the frontend interface and the database.

Several RESTful API endpoints were developed to support different functionalities of the tourism recommendation system. These endpoints include:

- User authentication APIs for signup and login
- Destination filtering APIs based on month and category
- Nearby attraction recommendation APIs
- Accommodation retrieval APIs
- Travel history storage and retrieval APIs

The backend server processes user inputs received from the frontend and performs necessary filtering and recommendation calculations before returning results to the client interface.

C. Database Management

The system uses a **SQLite relational database** to store

tourism data and user information. SQLite was selected due to its lightweight architecture, ease of deployment, and efficient integration with Node.js applications.

The database contains several tables that support the system's functionality, including:

Users Table

Stores user information such as user ID, name, email address, and password credentials used for authentication.

Destinations Table

Contains tourism destination records including destination name, city, state, country, geographic coordinates (latitude and longitude), destination category, descriptive information, best months to visit, and associated images.

History Table

Stores travel planning history for users, including selected travel month, interest category, destination name, and timestamp of the recommendation request.

Accommodations Table

Contains accommodation information for each destination, including hotel name, accommodation type, price range, distance from the destination center, images, and booking links.

This database structure allows the system to efficiently retrieve and process tourism data during the recommendation process.

D. Geospatial Distance Calculation

One of the core functionalities of the system is identifying nearby tourist attractions. This is achieved through geospatial analysis using the **Haversine distance formula**, which calculates the distance between two geographic coordinates based on latitude and longitude values.

The backend server retrieves geographic coordinates for the selected destination and calculates distances between that destination and other locations stored in the dataset. Destinations that fall within a predefined radius are identified as nearby attractions and included in the recommendation results.

This geospatial analysis enables the system to recommend additional tourist locations that can be visited within the same travel region.

E. AI-Based Travel Insight Generation

To provide additional contextual information for travelers, the system integrates an **AI-based travel insight module** using a generative AI service. The AI module generates travel recommendations for each destination, including:

- Suggested travel duration
- Recommended tourist activities
- Practical travel tips

These AI-generated insights enhance the information available to users and help them make more informed travel decisions.

F. Interactive Map Visualization

To improve user interaction and spatial understanding of destinations, an **interactive map visualization feature** was

implemented using the **Leaflet JavaScript library**. Leaflet allows the application to display geographic locations on an interactive map using markers positioned according to latitude and longitude coordinates.

When a user selects a destination, the map automatically displays the destination location and allows users to explore nearby attractions visually. This visualization helps users understand the geographic relationships between destinations and supports efficient travel route planning.

Through the integration of modern web technologies, geospatial algorithms, and artificial intelligence services, the implemented system provides a comprehensive platform for intelligent tourism recommendation and travel planning.

VI. RESULTS

The developed **AI-Enhanced Geospatial Tourism Recommendation System** was tested to evaluate its effectiveness in generating personalized travel recommendations and improving the overall travel planning process. The system was implemented as a web-based application and evaluated through experimental usage scenarios involving different travel preferences and seasonal conditions.

The primary goal of the evaluation was to verify whether the system could successfully generate relevant tourism recommendations based on user inputs such as travel month and destination category. The system was also evaluated to determine its ability to identify nearby tourist attractions and provide meaningful travel insights.

A. Destination Recommendation Performance

The destination filtering mechanism successfully generates recommendations based on the user's selected travel month and preferred travel category. When users select a travel month, the system filters the dataset to identify destinations that are suitable for visiting during that particular period. This ensures that recommended locations provide favorable weather conditions and appropriate travel experiences.

Similarly, the category-based filtering mechanism effectively identifies destinations that match the user's travel interests. For example, users who select the **"Beaches"** category receive recommendations for coastal destinations, while users selecting **"Heritage"** receive suggestions for historically significant locations.

The combination of seasonal filtering and category-based filtering significantly improves the relevance of recommendations compared to traditional keyword-based travel searches.

B. Nearby Attraction Identification

One of the key features of the proposed system is the ability to identify **nearby tourist attractions** based on geographic proximity. Using the Haversine distance calculation, the system determines the distance between the selected destination and other tourist locations stored in the dataset.

Destinations that fall within a predefined distance threshold are recommended as nearby attractions. This feature allows travelers to explore additional places of interest located within the same region, enabling more efficient travel planning.

For example, when a user selects a destination such as **Jaipur**, the system can identify nearby attractions such as **Amer Fort, Nahargarh Fort, and Hawa Mahal**, which are located within a short geographic distance. This feature helps travelers plan multi-location visits during a single trip.

C. AI-Generated Travel Insights

The integration of the AI-based travel insight module further enhances the usefulness of the system. For each selected destination, the system generates additional travel information including:

- Recommended travel duration
- Suggested tourist activities
- Practical travel tips

These AI-generated insights provide contextual information that helps travelers better understand the destination and plan their travel schedules more effectively. Instead of manually searching for travel guides or travel blogs, users can obtain useful information directly from the system interface.

D. Interactive Map Visualization

Another important feature of the system is the **interactive map visualization**, which displays tourist destinations using geographic markers. The Leaflet map interface allows users to visually explore the location of a destination and its nearby attractions.

Map visualization improves user engagement by allowing travelers to understand spatial relationships between tourist locations. It also provides a more intuitive way of exploring travel destinations compared to traditional text-based recommendation systems.

Users can zoom, pan, and interact with map markers to access detailed information about destinations.

E. User Experience and System Efficiency

The integration of modern web technologies ensures that the system provides a responsive and user-friendly interface. The React-based frontend allows dynamic rendering of destination recommendations and map elements, while the Node.js backend efficiently processes recommendation queries.

The SQLite database enables fast data retrieval for tourism information and filtering operations. Since the dataset is stored locally within the application environment, the system can generate recommendations quickly without requiring heavy external database queries.

Overall, the system demonstrates strong performance in terms of recommendation relevance, response time, and usability.

F. Comparative Advantages

Compared with traditional tourism platforms that rely mainly on keyword searches or popularity rankings, the proposed system provides several advantages:

- Personalized recommendations based on user interests
- Seasonal suitability filtering for travel planning
- Identification of geographically nearby tourist attractions
- AI-generated travel insights for destination guidance
- Interactive map-based exploration of tourist locations

- These features collectively enhance the travel planning process and provide users with a more intelligent tourism recommendation experience.

VII. CONCLUSION

This research presented an **AI-Enhanced Geospatial Tourism Recommendation System** designed to assist travelers in discovering suitable tourist destinations based on their preferences, travel timing, and geographic proximity between locations. The system integrates multiple technologies including user preference filtering, seasonal travel analysis, geospatial distance calculation, and AI-generated travel insights to provide meaningful and personalized travel recommendations.

The developed platform allows users to select their preferred travel month and category of interest, enabling the system to filter tourism destinations that match both seasonal suitability and personal interests. The integration of geospatial analysis using the **Haversine distance formula** enables the identification of nearby tourist attractions, allowing travelers to explore multiple locations within the same geographic region. This feature improves travel efficiency and supports better itinerary planning.

In addition to recommendation filtering and geospatial analysis, the system incorporates an **AI-based travel insight module** that generates contextual information such as recommended travel duration, suggested tourist activities, and practical travel tips. These insights enhance the overall travel planning experience by providing valuable guidance directly within the application.

The implementation of the system as a **full-stack web application** using React, Node.js, Express, SQLite, and Leaflet demonstrates the feasibility of integrating modern web technologies with geospatial analysis and artificial intelligence to develop an intelligent tourism recommendation platform. The interactive map visualization further improves user engagement by allowing travelers to explore destinations and nearby attractions spatially.

Experimental usage of the system shows that combining **preference-based filtering, seasonal analysis, geospatial proximity detection, and AI-generated insights** significantly improves the relevance and usefulness of tourism recommendations compared to traditional search-based travel planning platforms.

Although the proposed system demonstrates promising results, several opportunities exist for future improvements. Future work may focus on integrating **real-time data sources** such as weather information, traffic conditions, and crowd density analysis to further enhance recommendation accuracy. Additionally, the system could be extended to support **mobile applications and larger tourism datasets**, enabling broader scalability and accessibility for travelers.

Overall, the proposed AI-Enhanced Geospatial Tourism Recommendation System provides an effective solution for intelligent travel planning and demonstrates the potential of combining geospatial technologies and artificial intelligence to enhance tourism recommendation services.

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